

NeuroHome

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Abstract

In Portugal, many elderly people live alone in their homes at an advanced age. This situation not only causes safety problems, such as falls, but also mental health issues caused by advanced age and loneliness. Therefore, the objective of this work is to develop a system that not only improves home safety by detecting gas leaks and inadequate temperature, but also detects daily routines and their variations, to alert a caregiver to a possible health issue affecting the elderly person.

Keywords:

Ambient Assisted Living, Elderly Monitoring, Internet of Things, Non-invasive Sensors, Routine Learning, Mental Health Prevention.

Resumo

Em Portugal muitos idosos vivem sozinhos em suas casas, muitos deles com idade avançada. Isto não só causa problemas de segurança, como quedas, mas também problemas do foro mental provocadas por idade avançada e solidão. Assim, o objetivo deste trabalho é desenvolver um sistema que não só melhore a segurança na casa, detetando fugas de gás e temperatura inadequada, mas também que detete as rotinas e que detete as suas variações, de forma a avisar um cuidador sobre um possível mal-estar no idoso.

Palavras-chave

Ambiente de Vida Assistido, Monitorização da Terceira Idade, Internet e Integração, Sensores não Invasivos, Aprendizagem de Rotinas, Prevenção de Problemas de Saúde Mental.

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List of Software

Flutter / Dart: Cross-platform framework used to develop the caregiver mobile application.

MicroPython: Programming language for the ESP32 sensor node firmware and edge-level data processing.

Python: Primary language for the Raspberry Pi gateway operations and routine-learning algorithms.

PostgreSQL: Relational database management system used for storing sensor logs and historical data.

Wix: Web development platform utilized for creating the project's institutional website.

Microsoft Word: Documentation tool.

Chapter 1

Introduction

In this chapter it will be introduced the work and it will be stated its objectives and its motivation.

1.1 - Overview

The demographic landscape in Portugal reveals a significant socio-economic challenge, with the elderly population now exceeding **2.5 million individuals**. Notably, approximately **40% of citizens aged 80 and above reside in isolation**, a condition that drastically increases their vulnerability. This isolation presents two-fold risks: immediate physical dangers, such as domestic falls, and long-term psychiatric complications, including depression and cognitive decline triggered by loneliness.



Figure 1.1 - Elderly people

Current research and clinical testimonies suggest that **early detection of dementia** and other mental health conditions can be achieved through the monitoring of domestic movement, as severe changes in daily routines often serve as primary clinical warnings. However, most existing monitoring solutions are either prohibitively expensive or suffer from low user acceptance due to their invasive nature or the requirement for the user to wear hardware, such as pendants or bracelets, which are frequently forgotten.

1.2 - Motivation and Contents

The engineering motivation for this project is driven by the identified deficiencies in current telecare technologies. While various safety systems are commercially available, their implementation is often prohibitive by high acquisition costs and the requirement for professional installation, which limits accessibility for many families and community centres. Furthermore, existing solutions frequently fail to achieve high user adherence, as they tend to be perceived as intrusive or rely on manual maintenance, such as daily charging, that the elderly user may overlook.

A critical objective of this work is to pioneer a system that integrates physical safety with mental health monitoring. Current market alternatives, such as the **Vayyar Home**, focus predominantly on reactive fall detection. In contrast, **NeuroHome** is motivated by the potential to utilize domestic routine variations as a diagnostic tool for psychiatric conditions, including depression and early-stage dementia.

To ensure the viability of this solution, the development is guided by several core technical requirements:

- **Reliability and Efficiency:** The system must maintain stable communication across the domestic environment while ensuring low energy consumption to prolong battery life.
- **Data Integrity and Security:** One must ensure that all collected behavioural data is securely stored and protected to maintain the privacy and dignity of the user.
- **Operational Simplicity:** The interface for caregivers must present complex routine data in a clear, actionable format to facilitate early clinical intervention.

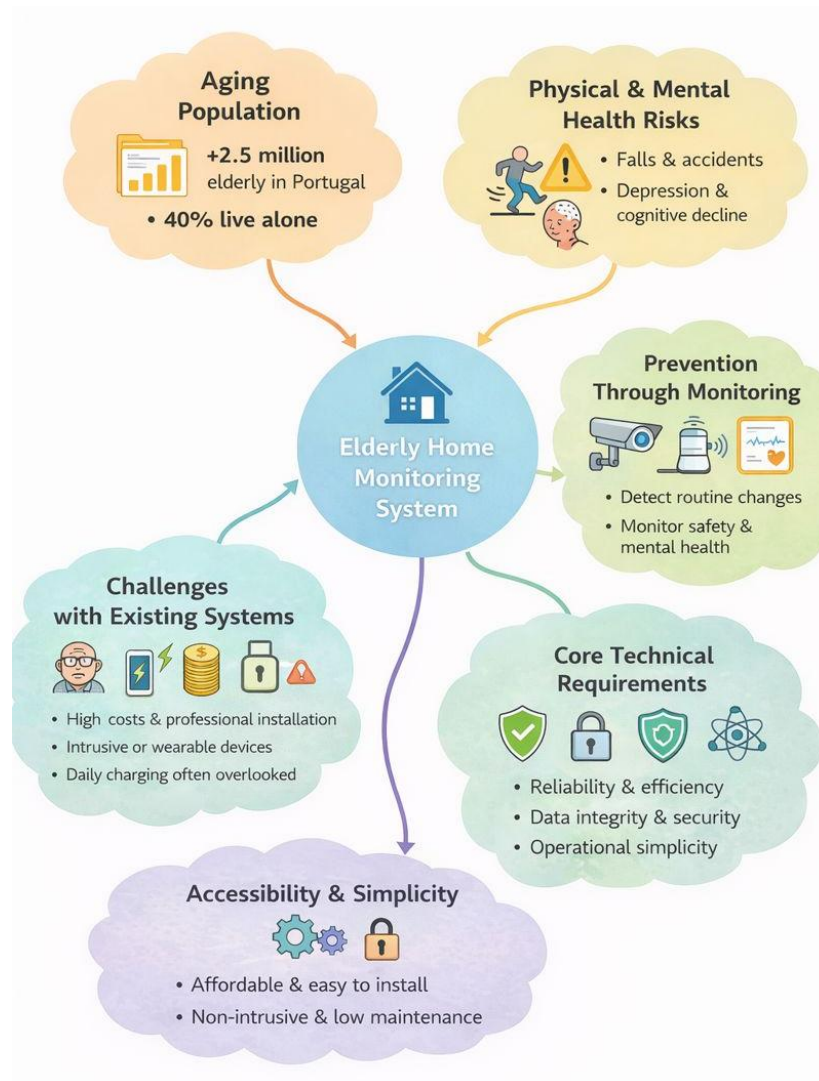


Figure 1.2 – Planning of the solution

Chapter 2

Overview of the solution (Planning phase)

This chapter talks about the conceptual solution, and how it is planned to achieve the desired objectives

The proposed solution is designed to monitor environmental safety and behavioural patterns within an elderly person's residence. The system aims to detect critical events, such as unsafe temperature fluctuations and gas leaks, alongside deviations in daily routines.

While environmental hazards are monitored via dedicated sensors, behavioural analysis is achieved by strategically positioning motion sensors at the different rooms. This configuration enables the system to track movement between zones, gradually constructing a profile of the user's daily habits.

Architecturally, the system follows a distributed sensing model. Peripheral nodes are installed in key living areas, specifically the kitchen, living room, bathroom, and bedroom. These nodes transmit data to a central gateway located in the hall. The gateway aggregates this information and uploads it to the cloud. Finally, a mobile application processes this data to visualize movement history and trigger real-time alerts for caregivers upon detecting environmental hazards or behavioural anomalies.

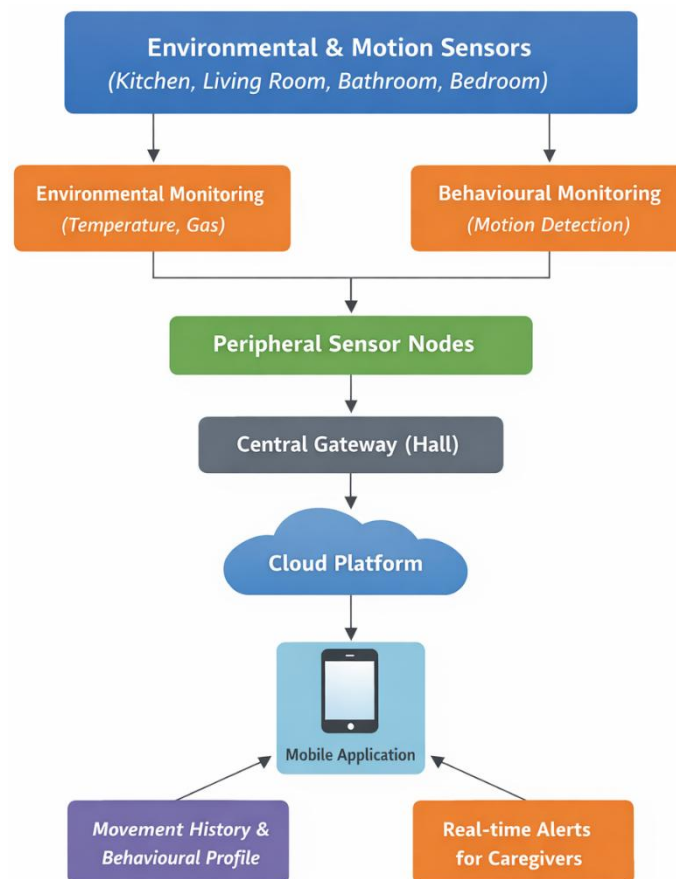


Figure 2.1 – Flowgraph of the conceptual solution

Chapter 3

Technical approach

(Do phase)

In this chapter it will be about the analyses of the technical part of the solution, system architecture and hardware integration.

3.1 - Data Acquisition Layer: Sensors

The monitoring system utilizes a multi-modal sensing approach to ensure environmental safety and behavioural tracking. Three distinct types of sensors are deployed across the residence to provide comprehensive data for the routine-learning algorithms.

3.1.1 - Temperature Monitoring

For thermal assessment, the **LM35DZ precision integrated-circuit temperature sensor** is employed. These units are strategically placed in the **bedroom** and **living room**, as these are the primary areas of occupancy where maintaining a stable and comfortable climate is critical for elderly well-being.

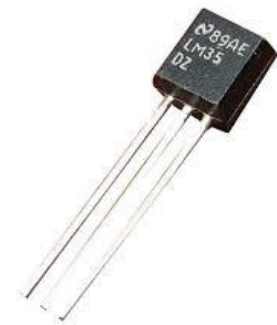


Figure 3.1 – LM35DZ sensor

- **Technical Rationale:** The LM35DZ was selected for its linear output scale and seamless compatibility with the ESP32's Analogue-to-Digital Converters (ADC).
- **Limitations:** While cost-effective, the sensor exhibits a precision and is limited to non-negative temperature ranges, which is sufficient for domestic indoor monitoring but may require calibration for high-accuracy applications.

3.1.2 - Occupancy and Transition Detection

Movement tracking is facilitated by **HC-SR04 ultrasonic sensors** positioned at the doorways between rooms.



Figure 3.2 – HC-SR04 sensor

- **Operating Principle:** These devices utilize high-frequency sound waves to measure distance based on Time-of-Flight (ToF) sensors. When a significant change in distance is detected, the system logs a transition event.
- **Performance:** The HC-SR04 offers a balance between low power consumption and high measurement precision, making it an ideal candidate for battery-powered peripheral nodes.

3.1.3 - Gas Detection

A **KEYESTUDIO KS0040 (MQ-2 based) sensor** is integrated into the kitchen node to mitigate risks associated with gas leaks or smoke.



- **Capabilities:** This sensor can detect a variety of combustible gases, including LPG, propane, and hydrogen.
- **Specifications:** Although highly sensitive and cost-effective, the sensor is non-selective, meaning it cannot differentiate between specific types of detected gases. It is calibrated to trigger a critical alert when concentrations exceed a safety threshold of **5000 ppm**.

Figure 3.3 – KS0040 sensor

3.2 - Processing and Communication Infrastructure

The system architecture follows a **distributed star topology**, comprising several peripheral nodes and a central gateway.

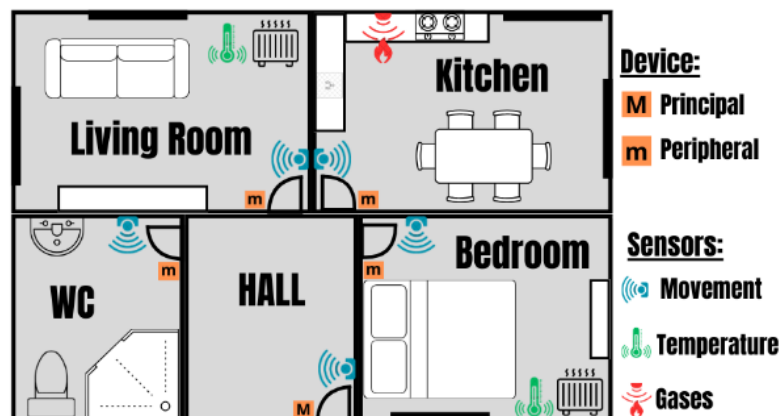


Figure 3.4 – Example of the system implementation

3.2.1 - Nodes (ESP32)

Each monitored room (excluding the hallway) is equipped with an **ESP32 microcontroller**.

- **Data Processing:** The ESP32 runs Python-based firmware (MicroPython) responsible for initial data acquisition and edge-level preprocessing. This includes filtering sensor noise and identifying outlier measurements before transmission.
- **Connectivity:** Data is transmitted to the central gateway via **Bluetooth communication**, optimizing the balance between range and energy efficiency. The ESP32's dual-core architecture allows for simultaneous sensor polling and communication without latency issues.

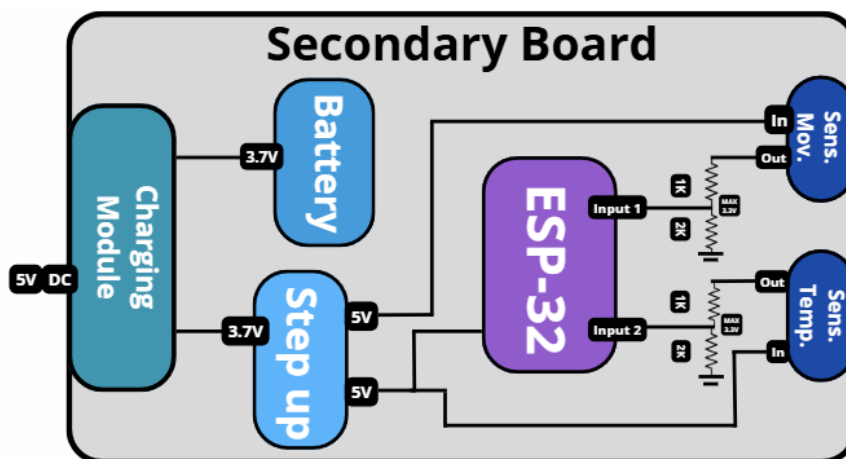


Figure 3.5 – Schematics of the peripheral board

3.2.2 - Central Gateway (Raspberry Pi 4)

A **Raspberry Pi 4** serves as the system's central hub, located in the entrance hall.

- **Primary Functions:** The gateway acts as a local server, aggregating data from all ESP32 nodes. It is also equipped with a dedicated movement sensor to monitor building entries and exits, which is vital for identifying wandering behaviours.
- **Cloud Synchronisation:** Utilizing its integrated Wi-Fi module, the Raspberry Pi uploads processed data periodically to the cloud. An onboard **SD card** provides local storage redundancy, ensuring data integrity during potential network outages.

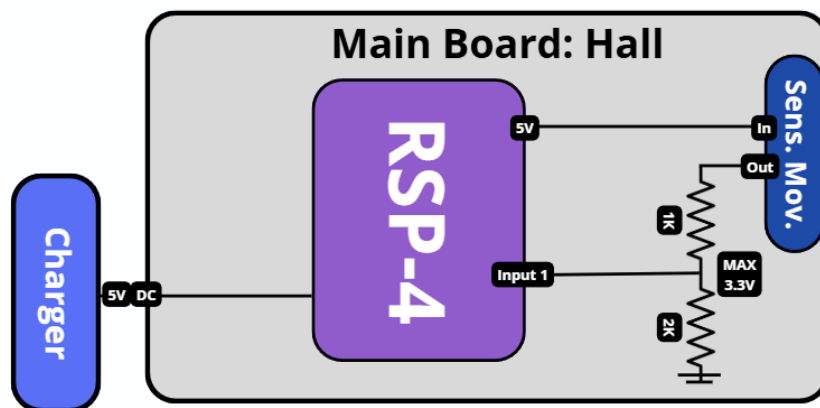


Figure 3.6 – Schematics of the main board

3.3 - Caregiver Application and User Interface

The interface layer was developed using the **Flutter framework**, providing a cross-platform mobile application for caregivers.

- **Functionality:** The application translates raw sensor data into a chronological movement history, allowing for long-term routine assessment.
- **Alerting System:** Real-time notifications are pushed to the user upon detection of environmental hazards (for example, gas leaks) or significant routine deviations (for example, excessive time spent in the bathroom).
- **System Controls:** To prevent false positives, the application includes a manual occupancy toggle, as the routine-learning logic is currently optimized for single-resident households.



Figure 3.7 – Homepage

3.3.1 - Real-time Occupancy and Notification Log

This interface facilitates the real-time monitoring of the resident's location by displaying the currently occupied division alongside a chronological log of room-to-room transitions. The screen integrates a notification system that provides immediate feedback on specific movement events, allowing caregivers to maintain continuous awareness of the elderly person's activity levels throughout the day. This data is essential for verifying immediate safety and ensuring that transition patterns remain consistent with expected daily behaviour.

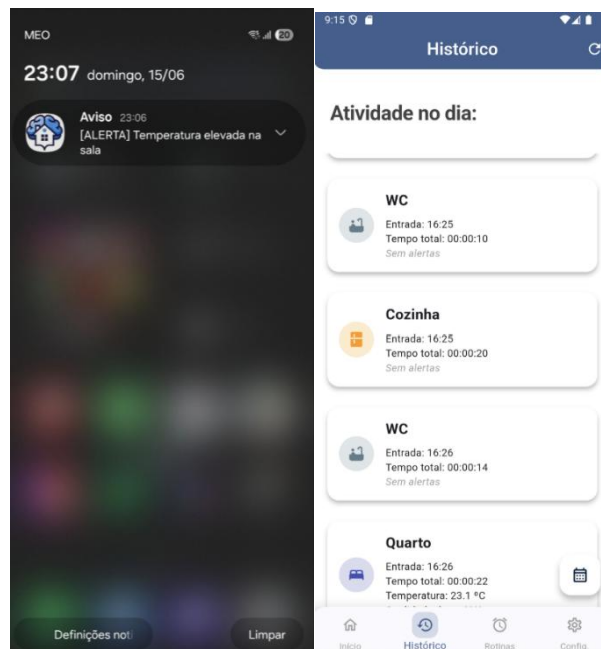


Figure 3.8 – Notification system and history of the day

3.3.2 - Diurnal and Nocturnal Routine Deviation Reports

The reporting module focuses on long-term behavioural analysis by comparing real-time data against a pre-established routine baseline. It generates distinct reports for diurnal and nocturnal cycles, specifically flagging significant deviations such as excessive periods of inactivity or unusual nocturnal wandering.

By highlighting these anomalies, the system provides clinical insights into potential health deteriorations or changes in cognitive state, enabling early intervention by healthcare providers.



Figure 3.9 – Routine Report

3.3.3 - Environmental Telemetry and Device Configuration

This section is dedicated to system management and the analysis of environmental safety metrics. It provides a detailed historical view of gas concentrations and temperature fluctuations for each monitored division, ensuring that safety thresholds are maintained. Furthermore, the interface allows for the technical configuration and status monitoring of the peripheral nodes (ESP32 units), ensuring the integrity and operational reliability of the entire sensing network.



Figure 3.10 – Configuration and room history

Chapter 4

Analyses of results

(Check and act phase)

In this chapter the results obtained after the system testing are going to be discussed.

4.1 – Validation Methodology and Pilot Phase

To assess the operational reliability and practical feasibility of the NeuroHome system, a pilot testing phase was conducted over a one week period. This stage was performed in collaboration with institutional partners, including Santa Casa da Misericórdia de Almada, Habicuidados, Community Centre of Laranjeiro and Rosana Baptista Apoio Domiciliário.

The primary objective of this validation was to evaluate the system's performance in a real-world domestic environment, focusing on the accuracy of transition detection, the latency of hazard alerts, and the overall acceptance of the non-invasive sensing model by the elderly users and their caregivers.

4.2 - Performance Analysis and System Efficacy

The quantitative results obtained during the pilot phase demonstrate the high efficacy of the proposed solution. The system achieved an **85% accuracy rate in room transition detection**, confirming the effectiveness of the ultrasonic sensor placement and the filtering algorithms implemented in the ESP32 nodes. Furthermore, safety-critical alerts for **gas leaks and temperature fluctuations exhibited a 90% success rate**, with a measured internal latency of approximately 1 second and a cloud to app notification time of roughly 10 seconds.

Regarding behavioural analytics, the routine-learning algorithm successfully identified standard patterns with a **70% precision rate** within the first week. While this metric is expected to improve with longer data exposure, it already allows for the identification of significant anomalies, such as prolonged inactivity.

The successful validation by professional partners confirms that NeuroHome provides a robust and low-cost alternative to invasive monitoring. The feedback indicates that the system's ability to operate without wearable hardware significantly improves user adherence, fulfilling the project's goal of enhancing safety while preserving the resident's dignity and privacy.

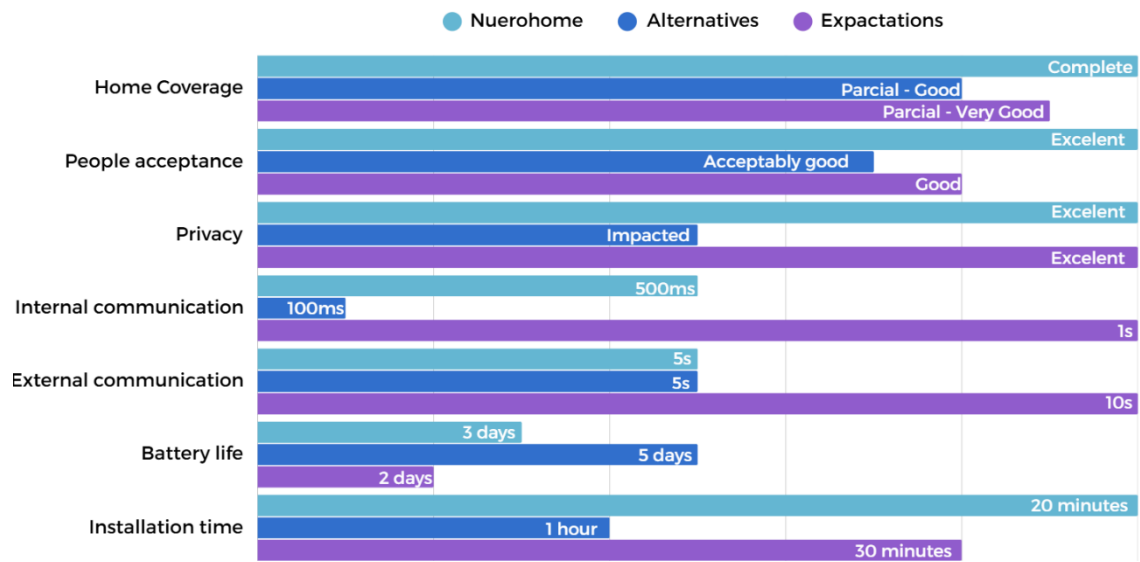


Figure 4.1 – Results obtained (1)

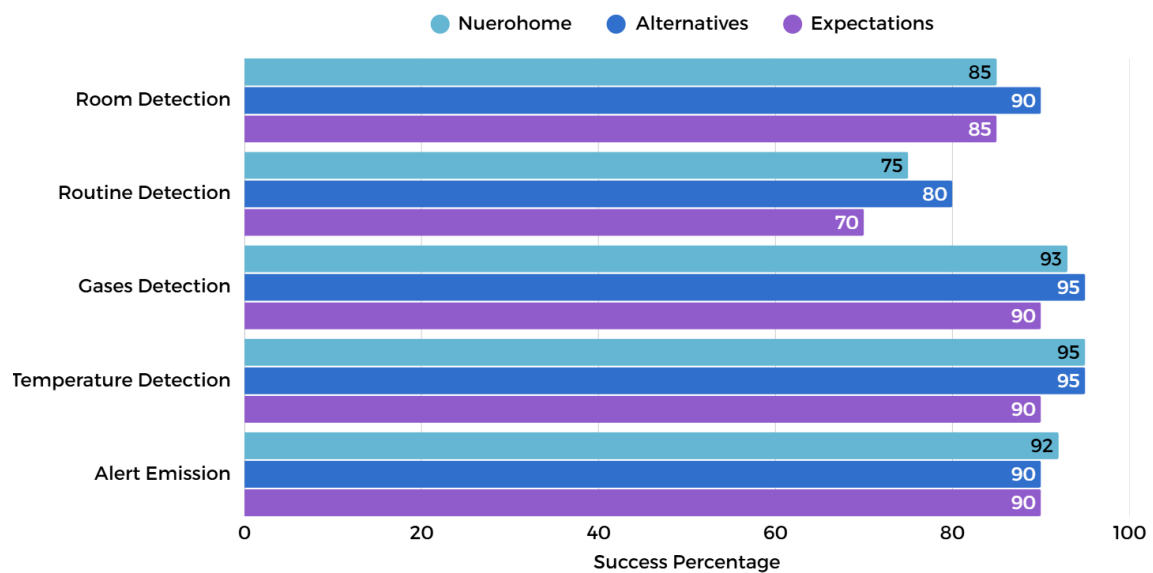


Figure 4.2 – Results obtained (2)

Chapter 5

Distribution of work

In this chapter it will be discriminated the contribution from every member in the development of this project.

The execution of the NeuroHome project was organized through in structured, ensuring that all technical, administrative, and communication requirements were addressed efficiently. The project was divided into three primary functional blocks: Data Acquisition (Hardware), Routine detection and processing (Software), and Mobile Application and system integration (Cloud/App).

While each member was assigned specific technical responsibilities, a cross-functional approach was adopted during the Check and Act phases, where collective monitoring and testing were performed to ensure system stability and integration.

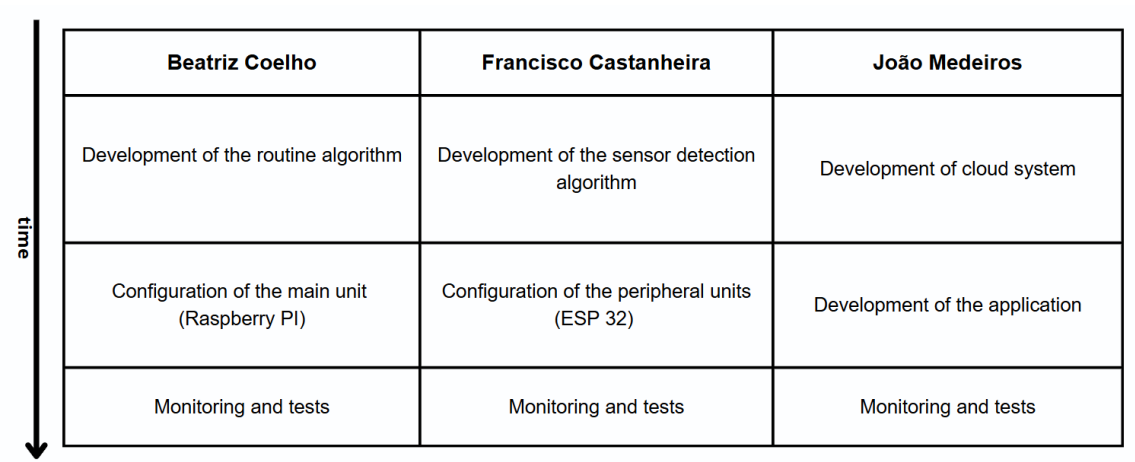


Figure 5.1 – Work distribution

Chapter 6

Conclusion

The **NeuroHome** project successfully demonstrated the feasibility of a low-cost, non-invasive system for monitoring the elderly in isolation. By achieving an **85% accuracy rate in room transition detection** and a **90% success rate in safety-critical alerts**, the prototype met its primary engineering objectives. The validation from institutional partners confirms that the system provides an alternative to invasive wearable hardware, effectively identifying behavioural anomalies without compromising user privacy.

As part of the **Act** phase, two critical areas for future development were identified: **energy optimization** and **algorithmic refinement**. To improve operational autonomy, future iterations must implement advanced deep-sleep protocols to extend the current 48 hours battery life of the peripheral nodes. Additionally, longer pilot phases will allow for more sophisticated models to enhance routine learning precision. Ultimately, NeuroHome offers a scalable solution to enhance the safety and well-being of the elderly population in Portugal.

Annexe 1

Minute from first meeting

Meeting Minutes 1: Project Kick-off and Planning (PLAN Phase)

Date: January 15, 2025

Time: 10:00 AM – 11:30 AM

Location: Online (Zoom)

Attendees: Beatriz Coelho, Francisco Castanheira and João Medeiros.

Absentees: None.

Agenda:

1. Project Planning and Objectives Definition.
2. Work Breakdown Structure and Task Assignment.
3. Preliminary Sensor Selection.

Discussion:

- **Project Goal:** The primary problem that the project should address is the isolation of the elderly in Portugal and the need for non-invasive monitoring, to combat the lack of follow-up during day and night by formal and informal caregivers.
- **Teams Needed:** It was decided to divide the project into three functional teams: Data Collection, Cloud/Processing, and Application.
- **Hardware:** A plausible option would be using ESP-32 devices for secondary boards due to low power consumption and built-in Wi-Fi. These devices can be installed in every room needed and operate via battery, making the device easy to hide.

Decisions:

- **Decision 1:** The Raspberry Pi 4 will be the central controller located in the hallway. This device has enough processing power to deal with all the data, will be plugged directly on a power supply (do not need to be as power efficient as the rest) and can properly communicate with all the ESP-32 devices, while be small enough to fit in a prototype box.
- **Decision 2:** The caregiver application will be developed to work on a mobile platform, being easily accessed and allowing the caregiver to access day and nights reports, to optimize reviewing the data collected.

Action Scheduling:

Activity	Description/Objective	Responsible	Deadline
Scope Definition	Detail isolation problems and non-invasive monitoring needs.	Whole Team	Completed (January 15)
Structuring	Divide the project into blocks: Data Collection, Cloud/Processing, and App.	Whole Team	Completed (January 15)
Literature Review	Research existing routine-learning algorithms. Evaluate medical criteria on health evaluation	Beatriz Coelho	January 30
Circuit Design	Prepare preliminary schematics for secondary boards (ESP-32). Evaluate data obtaining and processing.	Francisco Castanheira	February 05
App - UI	Create a visual and functional prototype of the caregiver application. Develop a cloudbased system to receive the data	João Medeiros	February 05

Adjournment: Meeting ended at 11:30 AM. The next meeting is scheduled for February 8, at 10:00.

Annexe 2

Minute from fourth meeting

Meeting Minutes 4: Progress Review and Technical Integration (DO/CHECK Phase)

Date: May 14, 2025

Time: 10:00 AM – 12:30 PM

Location: Online (Zoom)

Attendees: Beatriz Coelho, Francisco Castanheira and João Medeiros.

Absentees: None.

Agenda:

1. Hardware Prototype Testing Results.
2. Routine Algorithm Integration.
3. App-Cloud Communication Status.

Discussion:

- **Hardware Status:** Movement sensors are accurately detecting transitions between rooms within the 0.7m door-width range (success rate > 80%).
- **Algorithm Development:** Presented the first version of the routine-learning logic. There is a need to improve the precision of inactivity detection to avoid false alarms, which are happening very often.
- **App Integration:** Gas concentration alerts (above 5000ppm) are now being successfully pushed from the cloud to the mobile app. The goal now is to create, real time, alarms for long inactivity situations.
- **Main Goal Achieved:** The 3 modules from each team are working together, creating a functional prototype. Main features like gas and room detection are working, as well as the notifications and report generation.

Decisions:

- **Decision 1:** Battery life needs improvement (not lasting more than a day per charge – below expectations). The team will implement a sleep mode for ESP-32 when no movement is detected.
- **Decision 2:** Room detection history will be stored in the cloud database as well as all the report data needed.

Action Scheduling:

Activity	Description/Objective	Responsible	Deadline
Power Saving Implementation	Develop and implement sleep mode on the ESP-32 to increase battery life.	Francisco Castanheira	May 20
Algorithm Refinement	Adjust routine deviation thresholds to improve precision and avoid false alarms.	Beatriz Coelho	May 25
Application Finalization	Complete the visual layout and data presentation for the Daily Report in the application.	João Medeiros	May 25
Database Configuration	Set up the cloud system for local storage of room detection logs.	João Medeiros	May 25

Adjournment: Meeting ended at 13:00 PM. The next meeting will focus on final validation tests with project partners, and is expected to happen June 10, at 10:00 AM.

Annexe 3

Minute from fifth meeting

Meeting Minutes 5: Partner Validation and Final Refinements (CHECK/ACT Phase)

Date: June 10, 2025

Time: 10:00 AM – 12:30 PM

Location: Online (Zoom)

Attendees: Beatriz Coelho, Francisco Castanheira, João Medeiros.

Guests (Feedback provided via reports): Rosana Baptista (Rosana Baptista Apoio Domiciliário), Nuno Sampaio (Habicuidados), Sónia Valério (Santa Casa da Misericórdia de Almada).

Agenda:

1. Analysis of Pilot Test Results and Metrics.
2. Review of Feedback from Institutional Partners.
3. Implementation of Final Adjustments (Act Phase).
4. Preparation of Presentation Materials (Video, Website, Poster).

Discussion:

- **Pilot Results Analysis:** The team evaluated the success rates achieved during the testing phase: 85% for room detection and 90% for both gas and temperature alerts. Routine detection reached 70% precision, which was deemed acceptable for the current sophisticated algorithm version.
- **Institutional Partner Feedback:** Rosana Baptista confirmed that the system's ability to detect domestic movement is crucial for the early identification of dementia patterns. Nuno Sampaio highlighted the solution's superiority over wearable pendants, which seniors often forget to wear.
- **Technical Corrections:** Addressing Sónia Valério's feedback on power reliability, the team confirmed that the newly implemented sleep mode and battery management improved the secondary boards autonomy.

Decisions:

- **Decision 1:** The current 85% room detection accuracy is approved for the final prototype delivery.
- **Decision 2:** The caregiver application will include the "Daily/Nightly Report" format to better visualize routine deviations as tested.
- **Decision 3:** The final unit price will be set at €30 per room and €50 for the main unit to maintain the low-cost requirement.

Action Scheduling:

Activity	Description/Objective	Responsible	Deadline
Partner Feedback Integration	Review and document qualitative feedback from Rosana Baptista and Santa Casa partners.	Whole Team	Completed (June 10)
Final Sensor Calibration	Ensure gas and temperature sensors meet the 90% accuracy target for safety alerts.	Francisco Castanheira	June 20
Marketing Materials	Complete the NeuroHome website and the demonstration video.	João Medeiros	June 25
Routine Logic Refinement	Final polish of the routine detection algorithm based on trial data.	Beatriz Coelho	June 25
Final Report Submission	Compile all technical data, metrics, and individual contributions for final delivery.	Whole Team	July 1

Adjournment: Meeting ended at 12:30 PM. Final project submission is scheduled for July 4.

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